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MEAM 4600

https://github.com/jamehrotra/cellflow

# CellFlow: Modeling Single-Cell Differentiation Dynamics with Neural ODEs and Flow Matching

## 1. Introduction

A fundamental challenge in developmental biology is reconstructing how cells change over time. A stem cell does not instantly become a specialized blood cell or neuron. It passes through many intermediate states, gradually shifting which genes are active and which are silent. Single-cell RNA sequencing (scRNA-seq) enables measurement of gene activity across thousands of individual cells simultaneously, producing snapshots of cellular state at each time point. A typical experiment yields a matrix of roughly 10,000 to 20,000 cells, each described by expression levels across approximately 20,000 genes.

The core problem is that these snapshots are static. scRNA-seq destroys cells during measurement, so the same cell cannot be observed at multiple time points. Standard analysis tools such as Monocle3 and Scanpy infer developmental ordering through graph-based heuristics, but do not learn a generative model of the underlying dynamics and cannot simulate how cells change over time or predict the effects of genetic perturbations.

This project, CellFlow, treats cell differentiation as a continuous dynamical process and learns that process from data using four deep learning approaches applied to the Embryoid Body (EB) dataset: a direct Neural ODE on PCA space, a sequential VAE combined with a Neural ODE, a jointly trained VAE and Neural ODE, and a Flow Matching model. All methods are evaluated using Wasserstein distance in a shared 50-dimensional PCA coordinate space on held-out time points, enabling direct quantitative comparison across methods.

## 2. Dataset

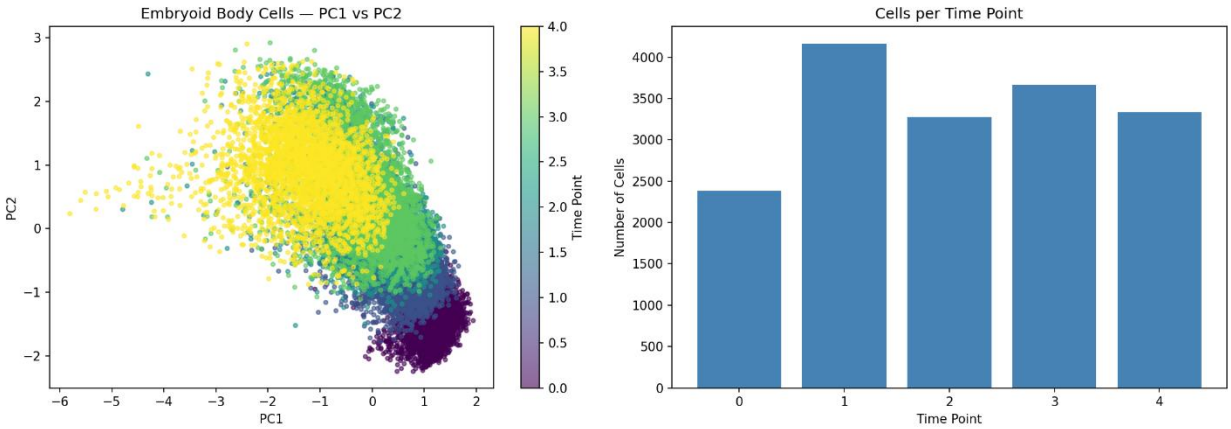



*Figure 1. EB dataset overview. Left: 16,819 cells projected onto PC1 vs PC2, colored by time point. Early cells (purple, t=0) cluster in the lower right; late cells (yellow, t=4) spread toward the upper left, showing learnable temporal structure. Right: cell counts per time point.*

All experiments use the Embryoid Body (EB) dataset from Moon et al. (2019), accessed via the TrajectoryNet package. The dataset consists of 16,819 mouse embryonic stem cells collected across five time points (t=0 through t=4), capturing differentiation from pluripotent stem cells toward multiple specialized lineages. The time points are reasonably balanced, with between 2,381 and 4,163 cells per snapshot.

The raw gene expression matrix was preprocessed using standard normalization and PCA. The TrajectoryNet package provides PCA embeddings at configurable dimensionality. Following an ablation study described in Section 5, all primary experiments use the top 50 principal components as input, standardized to zero mean and unit variance before training.

## 3. Methods

### 3.1 Neural ODE on PCA Space (Baseline)

The first approach learns a time-dependent vector field f(z, t) = dz/dt directly over the 50-dimensional standardized PCA space. The vector field is parameterized by a three-layer MLP with 64 hidden units and leaky ReLU activations, following the TrajectoryNet architecture. During training, cells from t=0 serve as initial conditions and are integrated forward using the Euler method via torchdiffeq. The training objective combines Maximum Mean Discrepancy (MMD) losses between predicted and observed cell distributions at each time point with an energy regularization term that penalizes large velocity magnitudes to prevent trajectory collapse. At evaluation, the dopri5 adaptive solver is used for more accurate integration.

### 3.2 Sequential VAE and Neural ODE

The second approach first trains a Variational Autoencoder (VAE) to compress the 50-dimensional PCA input into a 10-dimensional latent space, then fits a Neural ODE in that learned latent space. The VAE encoder and decoder each use two hidden layers of 128 units with batch

normalization, ReLU activations, and dropout of 0.1, following the scVI architecture. Training used KL annealing over 100 warmup epochs to prevent posterior collapse. The VAE identified 6 active latent dimensions through automatic relevance determination. The Neural ODE was then trained on those 6 active dimensions using the same MMD and energy regularization loss as the baseline.

### 3.3 Joint VAE and Neural ODE

Motivated by the failure of the sequential approach described in Section 4.2, a third method co-trains the VAE and Neural ODE simultaneously using a combined four-component loss: VAE reconstruction loss, KL divergence, dynamic regularization (MMD between ODE-predicted latent points and encoder-predicted latent points at each time point), and prediction loss (MMD between decoded ODE predictions and observed cells in PCA space). This approach is inspired by scNODE (Zhang et al., 2024) and addresses the key failure mode of the sequential pipeline by shaping the latent space to be consistent with ODE dynamics during training rather than after.

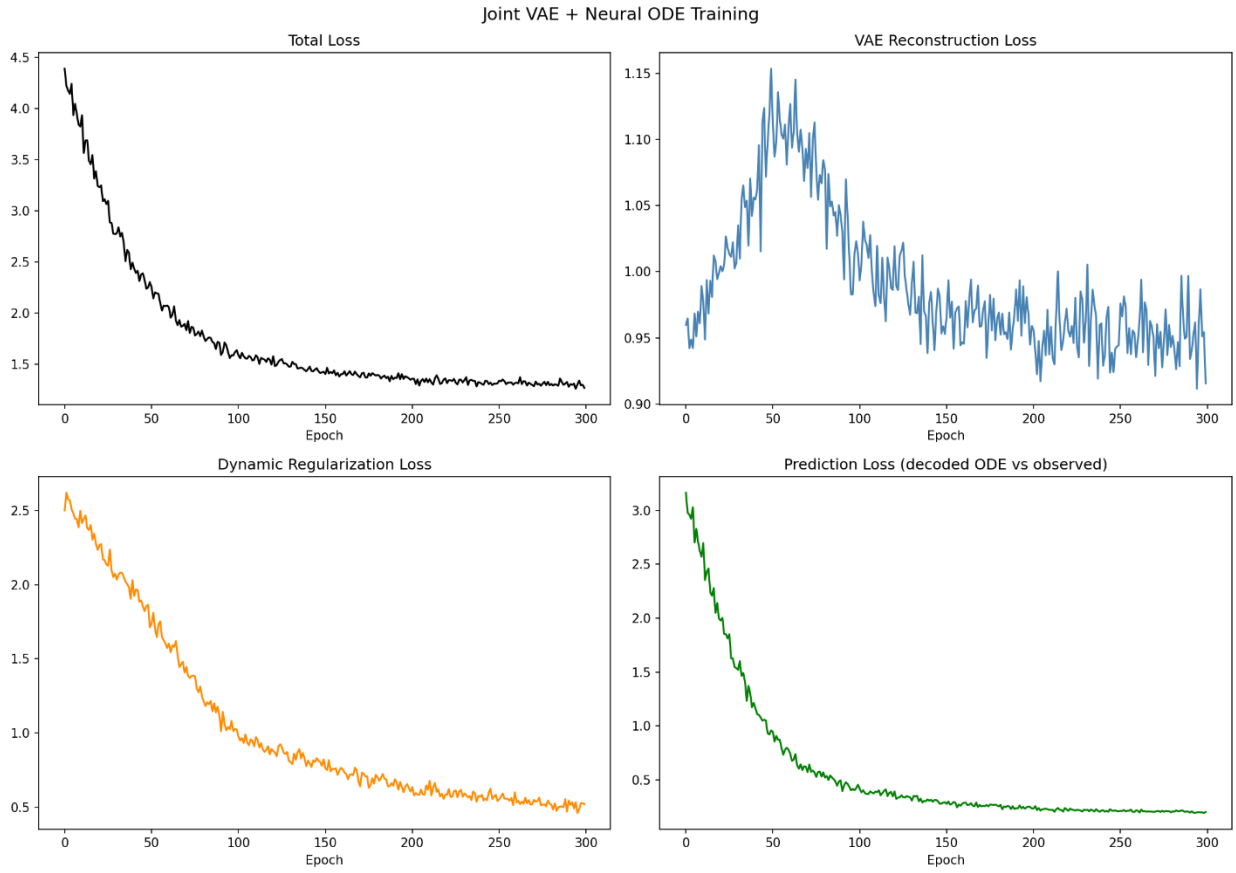



*Figure 2. Joint VAE and Neural ODE training curves. The dynamic regularization loss (orange) and prediction loss (green) both decrease steadily over 300 epochs, confirming that joint training successfully coordinates the VAE and ODE objectives.*

### 3.4 Flow Matching

The fourth approach applies Conditional Flow Matching (CFM) directly in the 50-dimensional PCA space. Rather than solving a differential equation during training, flow matching learns a velocity field v(z, t) that transports the source distribution of cells at t=0 toward target distributions at later time points by minimizing a simple MSE regression objective on interpolated cell pairs. The vector field is parameterized by a four-layer MLP with 64 hidden units and SiLU activations, implemented using TorchCFM. Training samples one random consecutive pair of time points per epoch. An improved variant using Optimal Transport CFM (OT-CFM) with 256 hidden units and cycling through all consecutive pairs per epoch was also evaluated and compared against the original.

## 4. Results, Failures, and Pivots

### 4.1 Failure: Initial VAE on 5-Dimensional Input

The original project design trained a VAE on the 5-dimensional PCA input, which is the TrajectoryNet default. This produced severe training instability. The KL loss and reconstruction loss were nearly equal throughout training, and the model failed to learn a meaningful latent representation. The root cause was architectural: compressing 5 dimensions to 4 provides almost no compression task, making the KL regularization term destabilizing relative to the reconstruction objective. This is consistent with the posterior collapse literature. The fix was to increase the PCA input dimensionality to 50, following MIOFlow which uses 200 PCA dimensions, giving the VAE a real compression task (50D to 10D) and producing a structured latent space with clear branching structure corresponding to known EB differentiation lineages.

### 4.2 Failure: Sequential VAE and Neural ODE Evaluation

An initial evaluation of the sequential VAE and Neural ODE appeared promising, showing low Wasserstein distances in the VAE latent space. However, this comparison was not scientifically valid. The two models were being evaluated in different coordinate spaces, which are not directly comparable. When both models were evaluated in the same shared 50D PCA space by decoding the VAE model's predictions back through the decoder, the sequential VAE and ODE approach was nearly four times worse (0.777 vs 0.224). The decoder introduces compounding errors when asked to reconstruct arbitrary latent points predicted by the ODE at unobserved time points, specifically points the decoder was never trained to handle. This critical finding motivated the joint training approach.

### 4.3 Failure: Trajectory Collapse Before Energy Regularization

Initial Neural ODE training without energy regularization produced collapsed trajectories. All predicted cells converged to a single point rather than spreading across the observed distribution. This is a known failure mode of Neural ODEs applied to single-cell data. The ODE finds a trivial solution that minimizes MMD by collapsing all cells toward the mean of the observed

distribution. Adding an energy regularization term that penalizes the squared L2 norm of the learned velocities resolved this issue, forcing the model to learn smooth, distributed paths through cell state space.

### 4.4 Failure: OT-CFM Did Not Improve Flow Matching

An improved Flow Matching variant was trained using Optimal Transport CFM with a larger network and a cycling training strategy. Despite producing a much cleaner training curve and lower MSE loss, the OT-CFM model performed worse on held-out evaluation (Wasserstein 0.472 vs 0.346). The random single-pair training strategy implicitly learns a more global velocity field, while OT-CFM with consecutive pair cycling appears to overfit to local transitions between adjacent time points and fails to generalize when integrating forward through the full trajectory.

### 4.5 Quantitative Results

Table 1 shows the final held-out Wasserstein evaluation averaged over five independent runs for all methods. All evaluations are in the shared 50D PCA coordinate space. Lower values indicate better performance.

| **Method** | **Avg Wasserstein (50D PCA)** | **Std** |
|---|---|---|
| Neural ODE on 50D PCA (direct) | 0.2275 | 0.0037 |
| Flow Matching (original) | 0.3457 | 0.0036 |
| Joint VAE + Neural ODE | 0.3854 | 0.0044 |
| Flow Matching (OT-CFM, improved) | 0.4717 | 0.0083 |
| Sequential VAE + Neural ODE | 0.7770 | 0.0036 |

*Table 1. Held-out Wasserstein evaluation over five runs. Lower is better.*

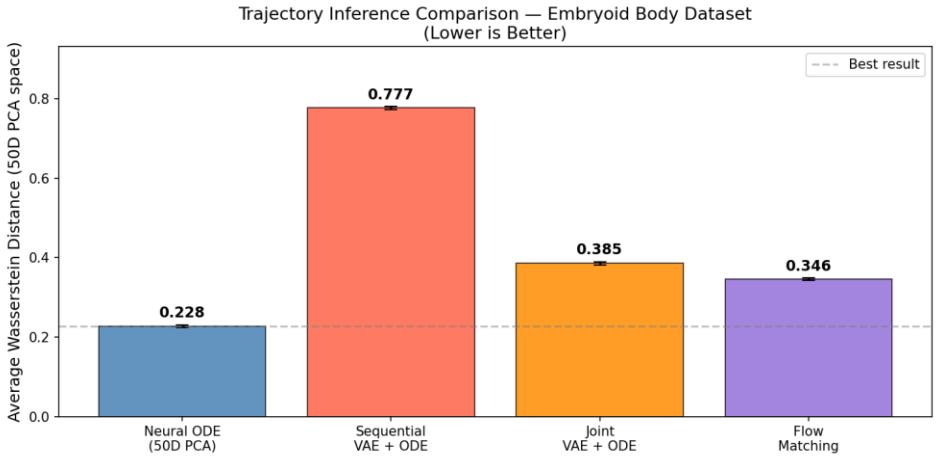


Figure 3. Final comparison of all methods in shared 50D PCA space. Error bars show standard deviation across five evaluation runs. Neural ODE on 50D PCA achieves the best performance.

## 4.6 Qualitative Analysis

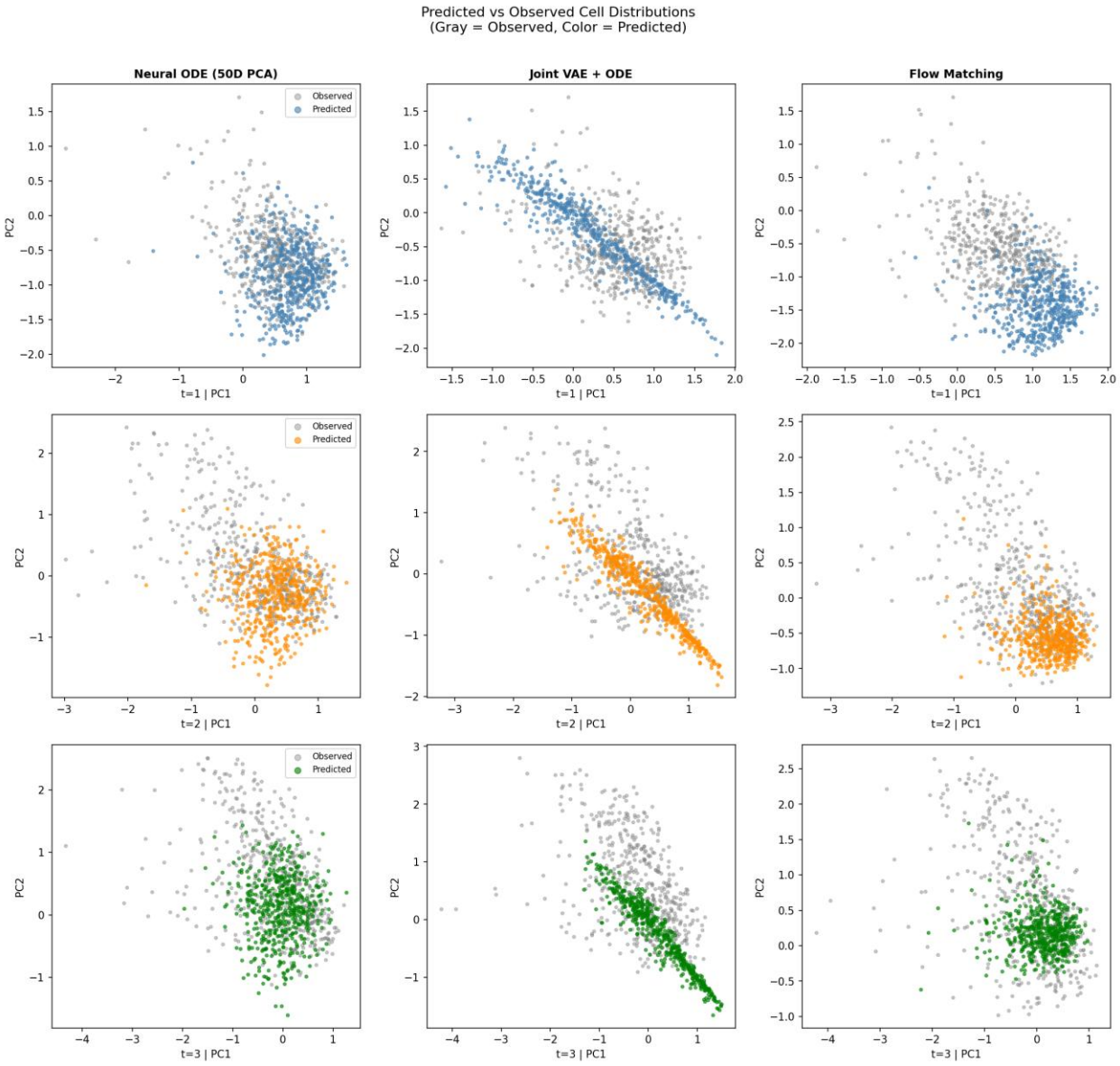



*Figure 4. Predicted (colored) vs observed (gray) cell distributions at t=1, t=2, and t=3 for the three best-performing methods. Neural ODE (left column) and Flow Matching (right column) produce diffuse, realistic distributions. Joint VAE + ODE (center column) collapses predicted trajectories to a narrow diagonal band at later time points.*

The qualitative comparison in Figure 4 reveals structure not visible in the Wasserstein numbers alone. The Joint VAE and ODE model, despite reasonable performance at t=1, collapses all predicted trajectories to a narrow diagonal manifold by t=2 and t=3. This suggests the dynamic regularization shaped the VAE latent space into a low-dimensional linear structure that constrains the ODE trajectories and reduces diversity in the predictions. Neural ODE and Flow Matching

both maintain realistic, diffuse distributions across all evaluated time points, better matching the spread of the observed cell populations.

## 5. Vector Field Visualization and Dimensionality Ablation

### 5.1 Learned Vector Field

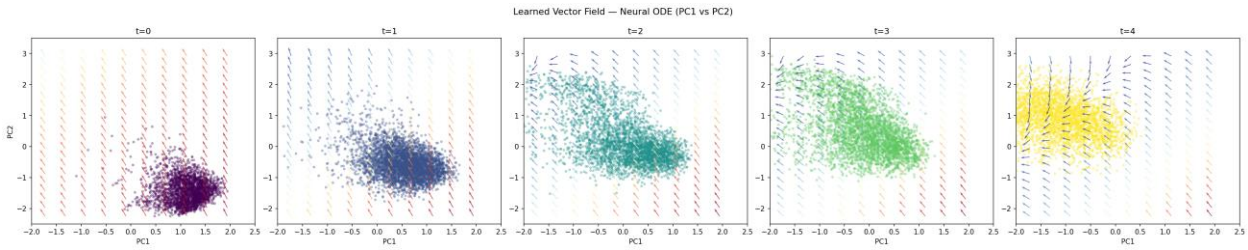



*Figure 5. Learned vector field of the Neural ODE at each time point (PC1 vs PC2). Arrow color indicates velocity magnitude (red = fast, blue = slow). The field shows converging dynamics at early time points and diverging, branching dynamics at t=2 and t=3, consistent with known EB differentiation biology.*

The vector field provides an interpretable view of what the Neural ODE learned. At t=0 and t=1, when cells are tightly clustered, the field converges toward the observed cluster, correctly pulling predicted cells toward the true distribution. At t=2 and t=3, the field develops a diverging, fan-like structure as the model captures branching differentiation characteristic of embryoid body development, where cells spread toward multiple lineages simultaneously. At t=4, the field shows a more complex pattern consistent with heterogeneous late-stage cell states.

### 5.2 Dimensionality Ablation

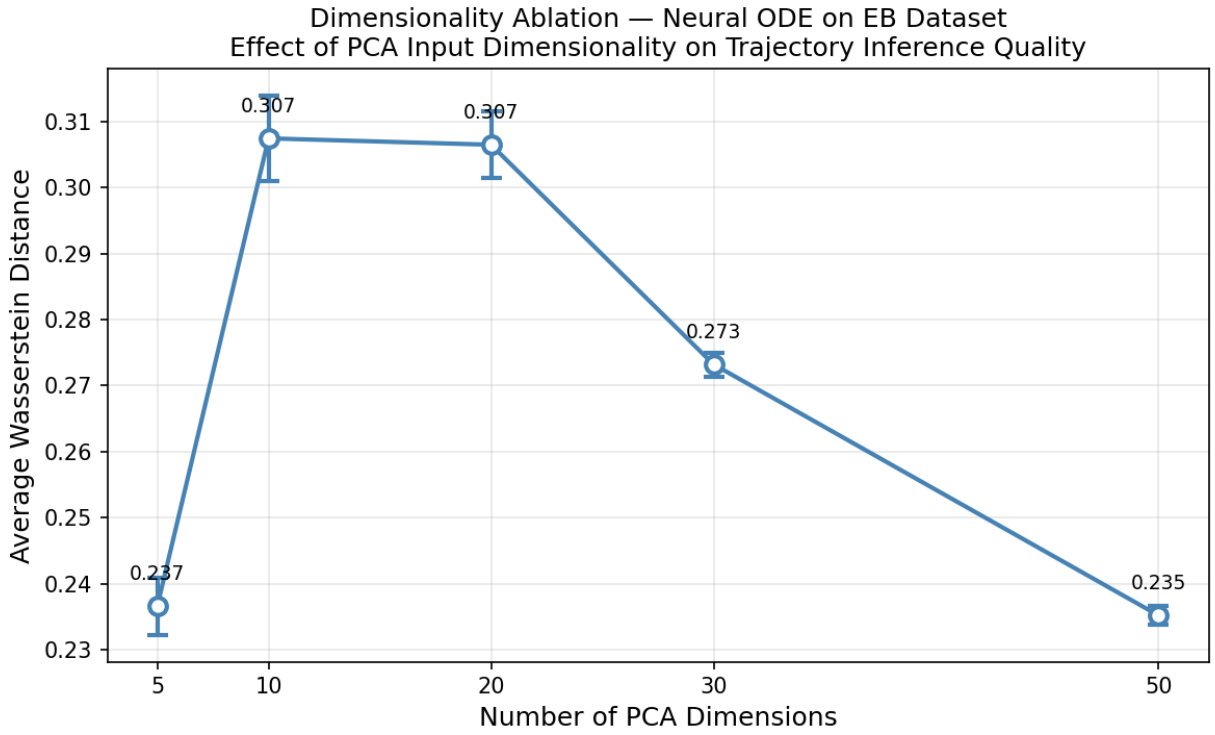



*Figure 6. Wasserstein distance as a function of PCA input dimensionality for the Neural ODE. Performance is non-monotonic, with intermediate dimensions (10D, 20D) performing worst and both 5D and 50D achieving similar accuracy.*

The dimensionality ablation reveals a non-monotonic relationship between PCA input dimensionality and trajectory inference quality. Intermediate dimensions (10D and 20D) produce the worst results, while both 5D (the TrajectoryNet default) and 50D achieve comparable accuracy. This pattern suggests the EB dataset has a low intrinsic trajectory dimensionality. The top 5 principal components appear to capture most of the trajectory-relevant information, and the degradation at intermediate dimensions likely reflects a transition zone where enough dimensions are present to introduce noise but not enough to capture the additional signal that higher dimensions provide. An important caveat is that Wasserstein distance scales with dimensionality, so the absolute values are not directly comparable across different dimensionality settings.

## 6. Discussion

The central hypothesis of this project was that nonlinear VAE compression would improve Neural ODE trajectory inference over raw PCA representations. The results do not support this hypothesis. The simple Neural ODE on PCA space (Wasserstein 0.228) outperformed all VAE-based approaches when evaluated fairly in a shared coordinate space. This finding partially contradicts claims from MIOFlow and scNODE that nonlinear representations improve trajectory

inference, though important differences in dataset scale and model sophistication limit direct comparison.

The key insight from this work is that VAE-based approaches introduce two compounding sources of error: the trajectory prediction error of the Neural ODE itself, and the reconstruction error of the VAE decoder when projecting predictions back to a common evaluation space. For the sequential pipeline, this compounding is severe (0.777). Joint training reduces but does not eliminate it (0.385). The decoder is not trained to handle arbitrary latent points at unobserved time points, and its generalization failure dominates the evaluation error.

Flow Matching achieves competitive performance (0.346) with considerably simpler and more stable training compared to Neural ODEs. There is no adjoint method, no ODE solver during training, and no risk of posterior collapse. The simulation-free training objective is easier to optimize and less sensitive to hyperparameter choices. That this simpler approach performs nearly as well as the best Neural ODE result while being substantially more practical to implement is a meaningful finding for practitioners.

Limitations of this work include the use of a single dataset, which limits generalizability. The EB dataset has a relatively clear temporal progression compared to many real biological systems. Datasets with more complex topology, sparser time points, or stronger batch effects may produce different relative performance across methods. Additionally, all evaluation was performed in PCA space rather than gene space, which may not capture all biologically relevant variation in predicted trajectories.

## 7. Conclusion

This project implemented and compared four deep learning approaches to single-cell trajectory inference on the Embryoid Body dataset. Several key findings emerged from the work. First, a direct Neural ODE on PCA space outperforms VAE-based approaches when evaluated in a shared coordinate space, because VAE decoders introduce compounding reconstruction error at unobserved time points. Second, sequential VAE and ODE pipelines fail substantially (Wasserstein 0.777) due to decoder generalization failure, and joint training only partially addresses this problem. Third, Flow Matching achieves competitive accuracy with substantially simpler training. Fourth, trajectory collapse is a consistent failure mode of Neural ODEs on single-cell data that requires explicit energy regularization to resolve. Fifth, the dimensionality ablation shows that 5D PCA performs similarly to 50D on this dataset, suggesting that the common default of using very few principal components does not significantly sacrifice trajectory inference quality on the EB dataset.

Future directions include jointly training VAE and Flow Matching components to potentially combine the benefits of both, applying these methods to larger and more complex single-cell

datasets, and incorporating RNA velocity information as an additional supervision signal for the learned dynamics.

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